CS242: System Software Lab

System Software Lab

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Outline

- Course, Attendance, Reference Book
- What do we study in this course?
- Why should this be studied?
- What is *"System Software Lab"* ?
- How is the course structured?

CS433 : Course site, Venue & Timing

- Course website: **http://jatinga.iitg.ernet.in/~asahu/cs242/**
- Tutorial Class
  - Venue : Core 5, Room- 5101
  - Timing: Mon 8.00AM-9.00AM
- Lab Class
  - Venue: B Tech II Year Lab, CSE, Ground Floor, Ext building
  - Timing: Tue 2.00PM-5.00PM

Course Structure

- Part I (Pre-midsem)
  - Overview of Unix system, commands and utilities;
    - Basic Linux administration and installation: grub, rpm, yum, disk partitioning; Basic Linux utilities, logging, backup, authentication;
  - Program maintenance: make, sccs, git
  - Debugging with gdb and ddd
- Part II (Post-midsem)
  - Archiving: shar, tar; shell use: redirection, .cshrc, environment variables;
  - Regular expression parsing: grep, egrep, sed, awk;
  - Shell programming: bash;

CS242 SysSoftLab: Text/Ref Books

- E. Nemeth, G. Snyder and T. R. Hein, Linux Administration Handbook, Prentice Hall PTR, 2002.
- S. Kochan and P. Wood, Unix Shell programming, 3rd Ed, SAMS, 2003.
- S. Das, Unix System V.4 Concepts and Applications, 3rd Ed, Tata Mcgraw-Hill, 2003.

CS343 : Grading and Rules

- Pre Mid Semester Part (46%)
  - Three Lab Evaluations: (7%+7%+7%)
  - Mid Sem Lab Examination: 25%
- Post Mid Semester Part (54%)
  - Three Lab Evaluations: (8%+8%+8%)
  - End Sem Lab Examination: 30%

Why should this be studied?

- Sys. Soft Lab : After CS101/110
- Why Linux?
  - Open Source, Command based, Driver easy
  - Better Structure
- Useful for designing system
  - Basic Sys Admin:
  - Programming Add: Make, GDB, VCS
  - Using Program useful: Shell, AWK/Sed

What do we study in this course?

- Useful for designing system
  - Basic Sys Admin: installing own software, user management, logging
  - Programming Add: Make, GDB, VCS/GIT
  - Using Program useful: Shell, AWK/Sed for File processing

Course Structure

- Part I (Pre-midsem)
  - A: Overview of Unix system, commands and utilities;
    - Basic Linux administration and installation: grub, rpm, yum, disk partitioning; Basic Linux utilities, logging, backup, authentication;
  - B: Program maintenance: make, sccs, git
  - C: Debugging with gdb and ddd
- Part II (Post-midsem)
  - A: Archiving: shar, tar; shell use: redirection, .cshrc, environment variables;
  - B: Regular expression parsing: grep, egrep, sed, awk;
  - C: Shell programming: bash;

Overview : Part I (A)

- Overview of Unix system, commands and utilities;
- Basic Linux administration and installation: grub, rpm, yum, disk partitioning;
- Basic Linux utilities, logging, backup, authentication;

Basic unix commands that everyone should know

(Even if you have a mac)

What the ~*&?!

~ “tilde” indicates your home directory: `/home/you`

* “star”: wildcard, matches anything

? wildcard, matches any one character

! History substitution, do not use

& run a job in the background, or redirect errors

#% special characters for most crystallography programs

` \ ([" ' back-quote, backslash, etc. special to shell

_ underscore, use this instead of spaces!!!

Where am I?

```
pwd
```

Print name of the “current working directory”

This is the default directory/folder where the shell program will look first for programs, files, etc. It is “where you are” in Unix space.

What is a directory?

`/home/yourname/whatever`

Directories are places you put files. They are represented as words connected by the "/" character. On Windows, they use a "\", just to be different. On Mac, they are called "folders". Whatever you do…

DO NOT PUT SPACES

In directory/file names!

What have we here?

`ls`

List contents of the current working directory

`ls -l` - long listing, with dates, owners, etc.

`ls -lrt` - above, but sorted by time

`ls -lrt /home/yourname/something`
- long-list a different directory

Go somewhere else?

cd

Change the current working directory

`cd /tmp/yourname/`
- go to your temporary directory

`cd -`
- go back to where you just were

`cd`
- no arguments, go back “home”

“home” is where your login starts

A new beginning…

`mkdir`

Create a new directory.

`mkdir ./something`	- make it
`cd ./something`	- go there
`ls`	- check its is empty

How do I get help?

`man`

Display the manual for a given program

`man ls` - see manual for the "`ls`" command
`man tcsh` - learn about the C shell
`man bash` - learn about that other shell
`man man` - read the manual for the manual

to return to the command prompt, type "`q`"

Move it!

mv

Move or rename a file. If you think about it, these are the same thing.

```
mv stupidname.txt bettername.txt
```

- change name

```
mv stupidplace/file.txt ../betterplace/file.txt
```

- same name, different directory

```
mv stupidname_*.img bettername_*.img
```

Will not work! Never ever do this!

Copy machine

`cp`

Copy a file. This is just like "`mv`" except it does not delete the original.

```
cp stupidname.txt bettername.txt
```

- change name, keep original

```
rm stupidname.txt
```

- now this is the same as "`mv`"

“Permission denied” !?

chmod

Change the “permission” of a file.

`chmod a+r filename.txt`

- make it so everyone can read it

`chmod u+rwx filename.txt`

- make it you can read/write/execute it

`chmod -R u+rw /some/random/place`

- make it so you can read/write everything under a directory

Destroy! Destroy!

`rm`

Remove a file forever. There is no “trash” or “undelete” in unix.

`rm unwanted_file.txt`

- delete file with that name

`rm -f /tmp/yourname/*`

- forcefully remove everything in your temporary directory.

Will not prompt for confirmation!

less is more

more

Display the contents of a text file, page by page

`more filename.txt` – display contents

`less filename.txt` – many installs now have a replacement for "`more`" called "less" which has nicer search features.

`head -n 5 filename.txt` – display 5 line contents

to return to the command prompt, type "`q`"

After the download…

`gunzip`

File compression and decompression

`gunzip ~/Downloads/whatever.tar.gz`

- decompress

`gzip ~/Downloads/whatever.tar`

- compress, creates file with `.gz` extension

Where the %$#& is it?

`find`

Search through directories, find files

`find ./ -name 'important*.txt'`

- look at everything under current working directory with name starting with "`important`" and ending in "`.txt`"

`find / -name 'important*.txt'`

- will always find it, but take a very long time!

Did I run out of disk space?

`df du`

Check how much space is left on disks

`df` - look at space left on all disks

`df .` - look at space left in the current working directory

`du -sk . | sort -g`

- add up space taken up by all files and subdirectories, list biggest hog last

Why so slow?

ps top

Look for programs that may be eating up CPU or memory.

`top` - list processes in order of CPU usage

`jobs` - list jobs running in background of current terminal

`ps -fHu yourname`

- list jobs belonging to your account in order of what spawned what

Die Die Die!

`kill`

Stop jobs that are running in the background

`kill %1` - kill job [1], as listed in "`jobs`"

`kill 1234` - kill job listed as 1234 by "`ps`" or "`top`"

`kill -9 1234` - that was not a suggestion!

`kill -9 -g 1234` – seriously kill that job and the program that launched it

Matching a specific pattern in a File

grep

- The **grep** tool searches for lines matching a specific pattern.
- The general form of the **grep** command is:

  grep pattern files
- Example:

```
$grep "Jon" *.c
 projaux.c: * Created by: Jon D. Smith
 projaux.c: * Last Modified by: Jon Smith
```

Manual

man

Manual about any command (all the in-out options and usage)

```
$man man
$man vim
$man gcc
$man cp
```

Merging two command

Unix pipe |

- pipe (|) creates a channel from one command to another.
- Think of the pipe as a way of connecting the output from one command to the input of another command.
- Examples:
  - Count the number of users logged onto the current system.
    - The **who** command will give us line by line output of all the current users.
    - We could then use the **wc -l** to count the number of lines...
    - **who | wc –l**

Logging

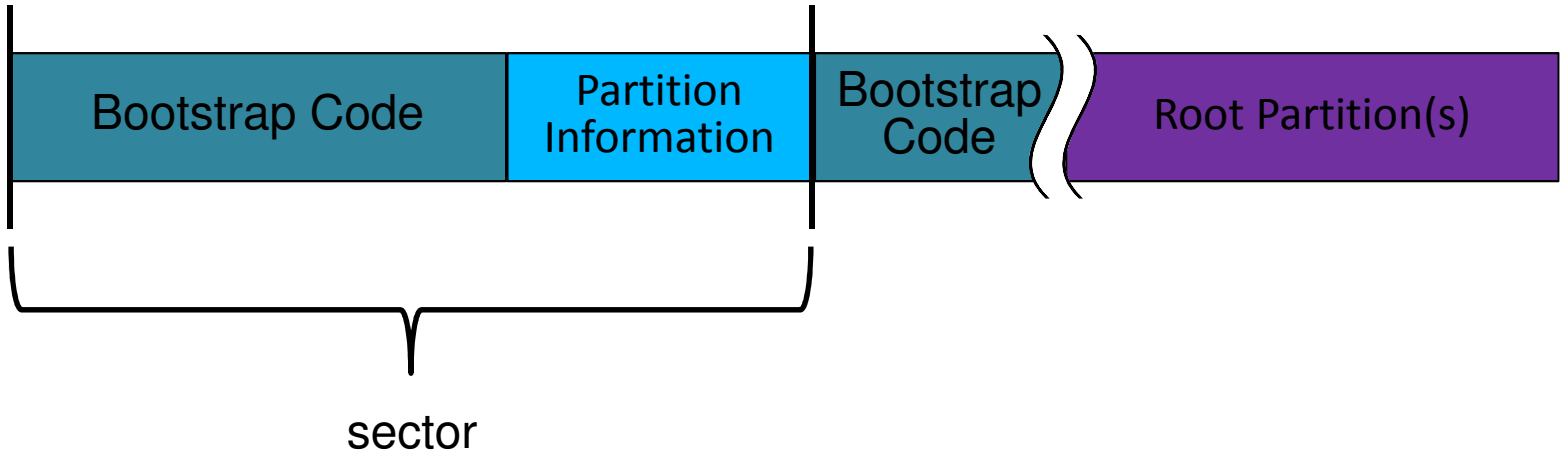
Linux Admin and installation

grub, rpm/yum, disk partitioning, and logging;

grub

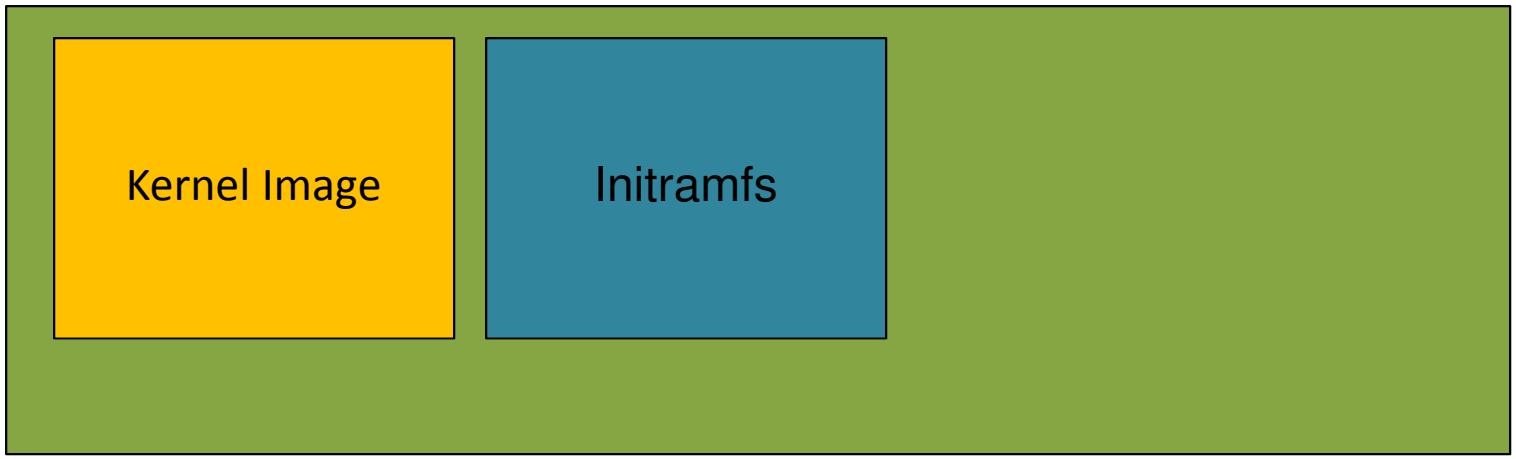
BIOS

- Used to install
  - Multiple OS in a system (Linux and Window)
  - Multiple version of same OS (many kernel of OS)
- Initialize the hardware
- Load the bootstrap code



Bootloader: grub

- Load the kernel
  - Image
  - Initramfs
  - Kernel command line parameters
- E.g., Grub



Linux Admin and installation

rpm, yum
or
dpkg or apt-get

Package management : Ubuntu Example

dpkg
apt-get

Debian package naming scheme:
package_version-build_architecture.deb

- ***package:*** name of the application being installed.
- ***version***: version number of the application.
- ***build:*** build number of the package. Each time the package is redone this number is incremented.
- ***architecture***: platform the package was compiled for.

There's a special type of package known as a ***task package***.
These packages are empty packages without software, but with dependencies. Used to install a large "task" on the system.

Using dpkg

Installing packages: ***dpkg -i package_file.deb***

```
debian:~# dpkg -i ethereal_0.8.13-2_i386.deb
Selecting previously deselected package ethereal.
(Reading database ... 54478 files and directories currently
installed.)
Unpacking ethereal (from ethereal_0.8.13-2_i386.deb) ...
dpkg: dependency problems prevent configuration of ethereal:
ethereal depends on libpcap0 (>= 0.4-1); however:
Package libpcap0 is not installed.
dpkg: error processing ethereal (--install):
dependency problems - leaving unconfigured
Errors were encountered while processing:
Ethereal
```

Using dpkg

Removing packages: ***dpkg --remove package_name.deb*** or ***dpkg -r package_name.deb***

```
debian:~# dpkg --remove libpcap0
dpkg: dependency problems prevent removal of libpcap0:
ethereal depends on libpcap0 (>= 0.4-1).
dpkg: error processing libpcap0 (--remove):
dependency problems - not removing
Errors were encountered while processing:
libpcap0
```

Using apt-get

installing packages: *apt-get install package_name*

```
debian:~# apt-get install nano
```

```
sofija@sofija-VirtualBox:~$ sudo apt-get install nano
Reading package lists... Done
Building dependency tree
Reading state information... Done
Suggested packages:
  hunspell
The following NEW packages will be installed:
  nano
0 upgraded, 1 newly installed, 0 to remove and 241 not upgraded.
Need to get 269 kB of archives.
```

disk partitioning

Using fdisk

debian:~# fdisk –l

debian:~# fdisk /dev/sdb2

m -> help

p -> print partition table

n -> create new partition

d -> delete partition

q -> quit without writing

w -> write to disk

Using fdisk

- On a disk, we can have a maximum of four partitions.
- Partitions are of two types.
  - Primary
  - Extended
- Extended partitions can have logical partitions inside it.
- Among the four possible partitions, the possible combinations are.
  - All 4 primary partitions
  - 3 primary partitions and 1 extended partition

Using fdisk

debian:~# mkfs.ext4 -j /dev/sdb1

Create ext4 FS in sdb1

debian:~# mkfs.fat /dev/sdb2

Create vfat in sdb1

Using gparted

debian:~# apt-get install gparted

GUI version of fdisk
More versatile

A document will be uploaded on gparted

Logging

- A Log file is a file that records events occurs in a operating system or other system software or the messages between the different user on a communication software.
- Logging is a act of keeping log files.

Significance 

System admin need to know what is happening on the system.

Logging

- A general purpose facility called syslog is used.
- Programme send their log entry to syslog by consulting two configuration files **/etc/syslogd.conf** and **/etc/syslog**.
- Syslog contain four basic terms:
  - **Facility:** Describe the type of application. For example: mail, kernel, ftp etc.
  - **Priority or Levels**: Describe the importance of log message. For example: emerg, crit, alert etc.
  - **Selector**: Combination of facility and levels.
  - **Action**: Whenever there is a match of selector, a action is performed eg: message to log file, echo the message to console.

Logging (Contd.)

- The syslog.conf controls where message are logged. A typical syslog.conf look like this:

```
*.err;kern.debug;auth.notice /dev/console
daemon,auth.notice           /var/log/messages
lpr.info                     /var/log/lpr.log
mail.*                       /var/log/mail.log
ftp.*                        /var/log/ftp.log
auth.*                       @prep.ai.mit.edu
auth.*                       root,amrood
netinfo.err                  /var/log/netinfo.log
install.*                    /var/log/install.log
*.emerg                      *
*.alert                      |program_name
mark.*                       /dev/console
```

Group Management

- Three type of account in the linux system:
  - **Root account**:Have complete control of system.
  - **System account**: Have some system specific control. For example: sshd account and mail account.
  - **User account**: Have interactive access to the system and provide very limited access to critical data.
- To manage user and groups, four type of administration files are needed:
  - /etc/passwd - Keep user account and password information.
  - /etc/shadow - Hold encrypted password.
  - /etc/group - contain group information.
  - /etc/gshadow - Hold secure information related to groups.

Thanks